Tutoring 1 - European Economic Policy

30-10-2026

Costanza Marconi
costanza.marconi@unibg.it

Exchange rate and competitiveness

E: nominal bilateral exchange rate

It tells the price of a foreign currency in terms of 1 unit of national currency

We use *E* when we want to compare the price of the same good i sold in two countries:

P^i_{US} *vs.* $P^i_{EZ} * E$

P^i_{EZ} *vs* P^i_{US} / E

Law of one price

If $P^i_{EZ} * E = P^i_{US}$, then also $P^i_{US} / E = P^i_{EZ}$ → $E = \frac{P_{US}}{P_{EU}}$

If the law of one price holds for all goods in a given basket, there is:

Purchasing Power Parity (PPP)

All countries' price levels are equal when measured in the same currency.
It means that I can buy the same good in both countries with the same amount of money. Let's check

Big Mac Index (July 2021)

Let's review:

Price Big Mac in EZ: €4.27
Price Big Mac in USA: $5.65
Market exchange rate: 1.18 < 1.32

What is the exchange rate necessary to obtain PPP?
Is it the same as the market exchange rate?
What are the implications?

E = 1.18

E for PPP: $E = \frac{P_{US}}{P_{EU}} = \frac{5.65}{4.27} = 1.32$

$P_{US} = \$5.65$ vs. $P_{EU} \cdot E = 4.27 \cdot 1.18 = \$5.04$

$P_{EU} = €4.27$ vs. $P_{US}/E = 5.65/1.18 = €4.79$

From PPP to real exchange rate

If PPP holds, then $E = P_{US} / P_{EZ}$

and also $P_{EZ}\, E / P_{US} = 1$

But PPP performs poorly in the real world

Deviations from PPP describe the actual exchange rate:

$E_R = P_{EZ}\, E / P_{US}$

From real exchange rate to measuring competitiveness

$E_R = P_{EZ}\, E / P_{US}$

If PPP holds, $E_R = 1$. In the real world, it can be:

$E_R > 1$ --> EZ goods are more expensive than US goods

$E_R < 1$ --> EZ goods are less expensive than US goods

What does it mean for competitiveness?

Competitiveness rate of change

Starting from $E_R = P_{EZ}\, E / P_{US}$, we are interested in change over time

From levels to rates of growth: how do we do it?

$$\Delta E_R = \frac{E_{Rt} - E_{Rt-1}}{E_{Rt-1}}$$

$$\Delta E_R = \Delta E \cdot \frac{\Delta P_{EU}}{\Delta P_{US}} \longrightarrow E_R = E \cdot \frac{P_{EU}}{P_{US}}$$

$$\log(E_R) = \log(E) + \log(P_{EU}) - \log(P_{US})$$

$$\Delta x = \frac{\partial \log(x)}{\partial t}$$

$$\frac{\partial \log(E_R)}{\partial t} = \frac{\partial \log(E)}{\partial t} + \left(\frac{\partial \log(P_{EU})}{\partial t} - \frac{\log(P_{US})}{\partial t} \right)$$

$\frac{\partial \log(E_R)}{\partial t} \rightarrow \Delta E_R$

$\frac{\partial \log(E)}{\partial t} \rightarrow \Delta E$

INFLATION DIFFERENTIAL

$\frac{\partial \log(P_{EU})}{\partial t} \rightarrow \Delta P_{EU}$ INFLATION €

$\frac{\log(P_{US})}{\partial t} \rightarrow \Delta P_{US}$ INFLATION $

Exercise A

Exchange rates:

$E_{t-1} = E_{\text{Sep 2024}} = 1.11618$ USD/EUR

$E_{t} = E_{\text{Sep 2025}} = 1.1736$ USD/EUR

Base price indices: $P^{EZ}_{t-1} = 100$ $P^{US}_{t-1} = 100$

Annual inflation rates (12-month to Aug 2025):

$\dot{p}^{EZ} = 2.0\%$ $\dot{p}^{US} = 2.9\%$

$P_{EU\,t} = 100\,(1 + 0.02) = 102$

$P_{US\,t} = 100\,(1 + 0.029) = 102.9$

$$\Delta P_{EU} = \frac{P_{EU\,t} - P_{EU\,t-1}}{P_{EU\,t-1}}$$

$$\Delta P_{EU} \cdot P_{EU\,t-1} = P_{EU\,t} - P_{EU\,t-1}$$

$$P_{EU\,t} = \Delta P_{EU} \cdot P_{EU\,t-1} + P_{EU\,t-1}$$

$$P_{EU\,t} = P_{EU\,t-1}\,(1 + \Delta P_{EU})$$

Comment on change in EZ-USA bilateral competitiveness over time by computing real exchange rates:

1. In levels

2. Using log / growth rate approximation

(1) LEVELS

$E_{R\,t-1} = ?$ $\quad E_{R\,t} = ?$ $\quad E_R = E \cdot \frac{P_{EU}}{P_{US}}$

$$E_{R\,t-1} = \frac{E_{t-1} \cdot P_{EU\,t-1}}{P_{US\,t-1}} = 1.11618 \cdot \frac{100}{100} = \underline{1.11618}$$

$$E_{R\,t} = E_t \cdot \frac{P_{EU\,t}}{P_{US\,t}} = 1.1736 \cdot \frac{102}{102.9} = \underline{1.1633}$$

$$\Delta E = \frac{E_t - E_{t-1}}{E_{t-1}} \qquad \Delta E_R = \frac{E_{R\,t} - E_{R\,t-1}}{E_{R\,t-1}}$$

$$\Delta E_R = \frac{1.1633 - 1.11618}{1.11618} = 0.0422$$

$4.22\,\%$

(2) RATES

$$\Delta E_R = \Delta E + \left(\dot{P}_{EU} - \dot{P}_{US}\right)$$

$$\frac{1.1736 - 1.11618}{1.11618} = 0.0514 = 5.14\,\%$$

$$\Delta E_R = 5.14 + 2 - 2.9 = \underline{4.24\,\%}$$

Exercise B

$\Delta E_n = ?$

$= -4.3\%$

Exchange rates:

E_{t-1} = 1.20 USD/EUR

E_t = 1.10 USD/EUR

Base price indices: P^{EZ}_{t-1}=100 P^{US}_{t-1} =110

Annual inflation rates:

$\dot{p}^{EZ}$=1% $\dot{p}^{US}$=4%